

ORIGINAL ARTICLE

Expression and functional analysis of G1 to S regulatory components reveals an important role for CDK2 in cell cycle regulation in human embryonic stem cells

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One of the characteristic features of human embryonic stem cells (hESCs) is the competence for self-renewal and pluripotency. To date, little is known about cell cycle regulation in these cells and how the cell cycle machinery influences hESCs properties. A common feature of human, murine and primate ESCs is the presence of a short G1 phase, which has been viewed as a time window during which stem cells are exposed to differentiation signals. We used the hESCs differentiation model and comparisons to human embryonic carcinoma (EC) cells to study the key regulators of G1 to S transition in hESCs. Our studies show that hESCs express all G1-specific CYCLINs (D1, D2, D3 and E) and cyclin-dependent kinases (CDK) (CDK2, CDK4 and CDK6) at variable levels. In contrast to murine ESCs, most of the cell cycle regulators in hESCs show cell cycle-dependent expression, thus revealing important differences in the expression of cell cycle regulatory components between these two embryonic cell types. Knockdown of CDK2 using RNA interference resulted in hESCs arrest at G1 phase of the cell cycle and differentiation to extraembryonic lineages. This suggests an important role for CDK2 in cell cycle regulation in hESCs that are likely to bear significant impacts on the maintenance of their pluripotent phenotype.

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Introduction

Human embryonic stem cells (hESCs) are derived from human pre-implantation embryos (Thomson *et al.*, 1998) and can proliferate without apparent limit, while retaining the ability to differentiate into multiple cell types (Carpenter *et al.*, 2003; Pera and Trounson, 2004; Stojkovic *et al.*, 2004a, b). Our understanding of the impact of cell cycle

structure on hESCs pluripotency is far from complete (Neganova and Lako, 2008) and is mostly derived from the extension of studies of mouse embryonic stem cells, which demonstrate substantial differences in cell cycle regulation between these and somatic cells (Stead *et al.*, 2002; Faast *et al.*, 2004; Savatier and Malashicheva, 2004; White *et al.*, 2005; Becker *et al.*, 2006; Neganova and Lako, 2008).

Primate ESCs seem to share some of these unusual cell cycle characteristics (Fluckiger *et al.*, 2006). Recent studies in hESCs suggest that these also possess an abbreviated cell cycle owing to a selective reduction of G1 phase (Becker *et al.*, 2006), which has been attributed to the elevated mRNA levels of CYCLIN D2/cyclin-dependent kinase 4 (CDK) that are known to promote G1 progression and to very low levels of inhibitory cell cycle proteins, p21, p27 and p57 (Becker *et al.*, 2006, 2007).

It is likely that the duration of each phase of the cell cycle of hESCs has a function in maintaining pluripotency and self-renewal; however, this has not been studied in detail. A link between cell cycle length and the differentiation process has been established in cancer cells, somatic cell lines and adult stem cells. For example, artificial lengthening of the cell cycle by inhibition of cyclin-dependent kinases can induce neural progenitor cells to form neurons (Calegari and Huttner, 2003) and shortening of the cell cycle of neural progenitor cells may prevent their differentiation (Calegari *et al.*, 2005). These observations suggest that G1 phase corresponds to a window of increased sensitivity of stem cells to differentiation signals.

In this study, we present data describing the expression of the main cell cycle components involved in G1 to S transition in hESCs and show that in contrast to murine ESCs, most of the cell cycle regulators in hESCs show cell cycle-dependent expression and kinase-associated activities. We also present functional data that support a key role for CDK2 in cell cycle regulation in hESCs.

Results

Expression analysis for key cell cycle regulatory components in hESCs

To analyse the expression of key cell cycle regulatory components in hESCs, we used flow cytometry-based assays (Figure 1a), direct immunoblotting (Figures 1b and c)

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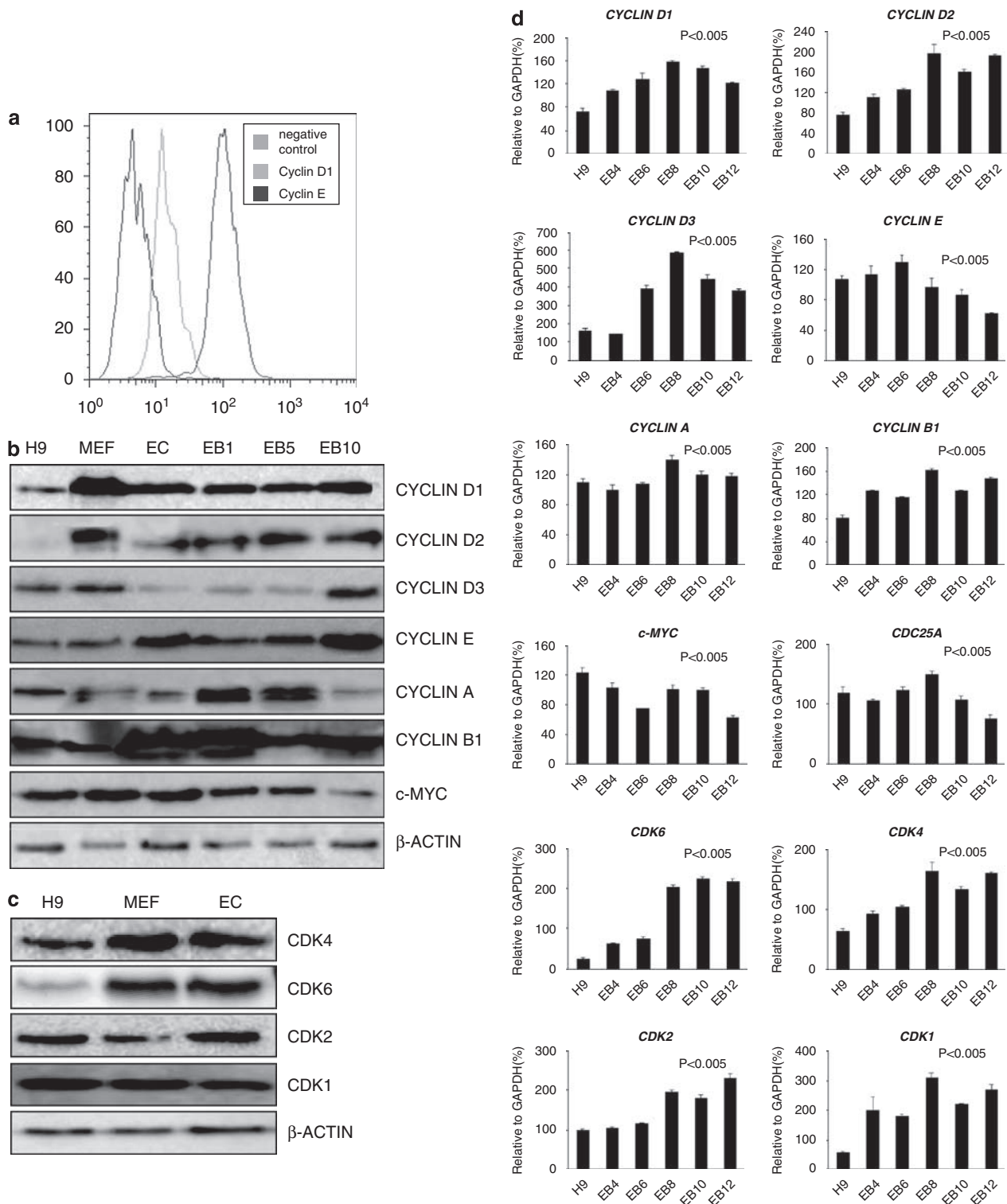


Figure 1 Expression analysis of main CYCLINS and CDKS in hESCs and during the course of their differentiation. **(a)** Flow cytometry analysis of expression of CYCLIN D1 and CYCLIN E at hESCs. **(b)** Expression analysis of CYCLINS and c-MYC evaluated by immunoblot analysis of extracts (15 µg protein) prepared from unsynchronized hESCs (H9), MEF and EC cells, and during the course of hESCs differentiation towards EBs. EB 1, 5 and 10-embryoid bodies at days 1, 5 and 10 of differentiation. β-ACTIN was used as loading control. This is a representative example of five independent experiments carried out in both hESC lines. **(c)** Immunoblot analysis of CDK4, CDK6, CDK1 and CDK2 in hESCs (H9), MEF and EC cells. β-ACTIN was used as loading control. This is a representative example of five independent experiments carried out in both hESC lines. **(d)** Quantitative RT-PCR analysis of CYCLINS, CDKS, c-MYC and CDC25A in hESCs and during their differentiation. Results are presented as average ± s.d. (n = 3). Statistical significance was assessed using ANOVA two factors. EB, embryoid body; hESCs, human embryonic stem cells; MEF, murine embryonic fibroblasts.

and quantitative RT-PCR (Figure 1d). The flow cytometry analysis demonstrated that the majority of cells express both CYCLIN D1 and E; however, CYCLIN E seems to be expressed at higher levels (Figures 1a and d). Quantitative RT-PCR showed that all D-type cyclins (*CYCLIN D1*, *D2* and *D3*) are expressed in hESCs (Figure 1d). These results are supported by direct immunoblotting that revealed comparable CYCLIN D1 and D3 expression in hESCs and very low to almost undetectable levels of CYCLIN D2, which can be because of the sensitivity of the antibody (Figure 1b). During the course of hESCs differentiation, the protein level of CYCLIN D1 and D2 increased steadily but the protein level of CYCLIN D3 was maintained at very low levels during the first 5 days of differentiation (Figures 1b and d) and a significant increase was seen at day 6 of differentiation. An increase in expression during hESCs differentiation was also noted for CYCLIN B1 (Figures 1b and d). In contrast, c-MYC showed an overall decrease in expression during the course of differentiation. Both the quantitative RT-PCR and immunoblotting showed fluctuations in CYCLIN E, CDC25A and CYCLIN A expression during hESCs differentiation (Figures 1b and d). Embryonic carcinoma (EC) cells showed higher expression of CYCLIN D1, D2, E, B1 and c-MYC, and lower expression of CYCLIN D3 and A when compared with hESCs (Figure 1b).

Next, we examined the expression of CDKs in hESCs, EC and murine embryonic fibroblast cells (Figures 1c

and d). All four CDKs were expressed in both hESCs and EC cells; however, CDK6 seemed to be less abundant in hESCs when compared with CDK1, 2 and 4. In addition, higher expression of CDK4, CDK6 and CDK2 and lower expression of CDK1 was noted in EC cells when compared with hESCs. Quantitative RT-PCR also indicated a significant increase in expression of all CDKs during hESCs differentiation (Figure 1d). Direct immunoblotting revealed different results to the quantitative RT-PCR, with CDK2 being downregulated during hESCs differentiation, and CDK6 and CDK4 being markedly upregulated (Supplementary Figure 3). CDK1 levels were maintained at more or less the same level during the differentiation process (Supplementary Figure 3).

To analyze the association between D-Cyclins and their catalytic partners and kinase activities of those complexes, we performed immunoprecipitation studies followed by kinase assays (Figures 2a and b). Significant amounts of CYCLIN D1 were found in complexes with CDK6 and to a lesser extent with CDK4 (Figure 2a). Also, CYCLIN D3 was found to be associated with CDK4 and CDK6 (Figure 2a). In contrast, CYCLIN D2 was mostly found in association with CDK4 and very small amounts co-immunoprecipitated with CDK6. Formation of complexes important for further progression through G1 phase was confirmed by immunoprecipitation studies, which revealed the presence of a CDK2 and CYCLIN E complex as well as CDK2-CDC25A (Figure 2a). CDC25A was also

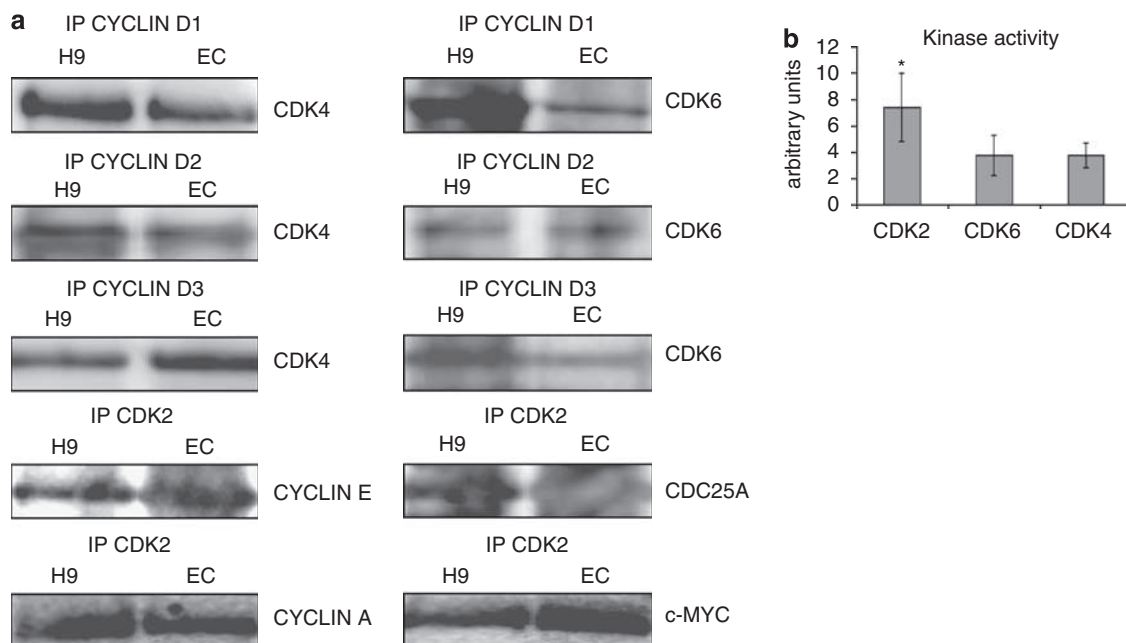


Figure 2 Analysis of specific CYCLIN-CDK complexes in hESCs and EC cells. (a) Immunoprecipitation analysis of the whole cell lysates (400 µg protein) from hESCs (H9) and EC showing formation of complexes between CYCLIN D1, CYCLIN D2, CYCLIN D3 and their corresponding kinases such as CDK4 and CDK6, as well as complexes between CDK2 and CYCLIN E, CDK2-CDC25A, CDK2-CYCLIN A and CDK2-c-MYC. In all panels a representative example of at least three independent experiments is shown. A no antibody control was included in all experiments (data not shown). All the reverse immunoprecipitations were performed followed by direct immunoblotting to confirm the specificity of interactions (data not shown). (b) Comparison of CDK6, CDK4 and CDK2-associated kinase activity in hESCs using RB as a substrate. Results are presented as average $\pm$ s.d. ($n = 3$). Statistical significance was Student's *t*-test (* $P < 0.05$). EC, embryonic carcinoma; hESCs, human embryonic stem cells.

associated with CDK4 and CDK6–CYCLIN-associated complexes (data not shown). Also, a strong association between CDK2 and CYCLIN A, as well as CDK2 and c-MYC was revealed (Figure 2a). When the activities of CDKs, known to be important for G1 to S phase progression, were compared with each other, the highest activity was observed for CDK2 (Figure 2b), which also correlates to highest CDK2 expression in hESCs compared with CDK4 and CDK6 (Figure 1c). All the immunoprecipitation and kinase activity assays were performed at the same time for all three kinases and normalized per unit of protein. They were, however, not optimized for specific CDKs (for example, CDK4 shows sensitivity to detergents that has not been noticed for CDK2 and CDK6), thus it is possible that these results might not reflect the true comparison of the activity of these CDKs *in vivo*. In addition, these assays were carried out in unsynchronized cell populations; hence, comparison of activity of each CDK was further investigated in synchronized cell population as shown in Figure 3d.

Expression analysis of G1/S regulatory components during cell cycle progression in hESCs

Synchronization experiments combined with mRNA and protein expression analysis were carried out to investigate fluctuations in expressions of different types of CYCLINS and CDKs during cell cycle progression. We used the nocodazole–aphidicholine synchronization procedure to collect cells in G1 and nocodazole alone to collect cells in G2 phase of the cell cycle as described in Materials and methods section and shown in Figure 3a. To obtain cells in S phase, hESCs were synchronized in G2 phase by incubation with nocodazole for 16 h and then after five extensive washes with media, the cells were released from the inhibitor and allowed to proceed along the cell cycle. Maximal percentage of cells in S phase was obtained after 9 h after release from nocodazole block (Figure 3a). These synchronization strategies have been used successfully for hESCs cycle analysis (Ghule *et al.*, 2007).

Quantitative RT–PCR analysis of hESCs populations synchronized in G1, S and G2 stages revealed a cell cycle-dependent expression for all key regulatory cell cycle components (Figure 3b). We noted that *CDK6*, *CDK2*, *CYCLIN D1* (Figure 3b) and *CYCLIN D3* (data not shown) showed the highest expression at G1, *CYCLIN E*, *CDK1*, *CYCLIN A*, *CYCLIN B1*, *c-MYC* (Figure 3b) and *CYCLIN D2* (data not shown) at G2, *CDC25A* at S phase and *CDK4* at G1 and S phases (Figure 3b). In addition to transcriptional control, ubiquitin-mediated degradation, post-translational modifications and cellular sublocalization can influence the expression of cell cycle regulatory components. In view of this, we conducted direct immunoblotting, although we recognized that the quantitation sensitivity will be much lower than quantitative RT–PCR. This analysis revealed that *CYCLIN D1*, *CDK6*, *CDK1*, *CDK4* (Figure 3c), *CYCLIN D2* and *CYCLIN D3* (data not shown) protein expression was maintained at

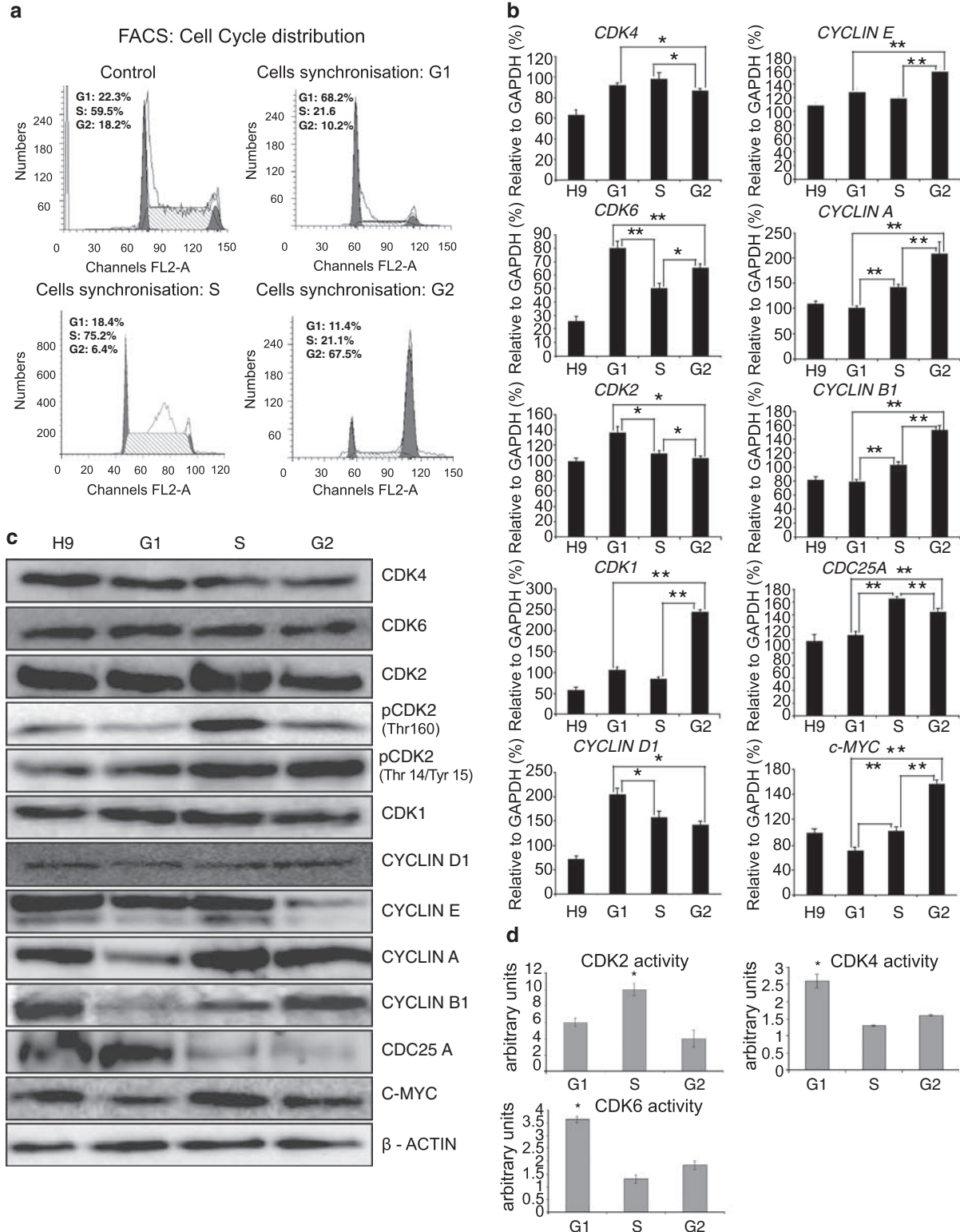
more or less constant level during the cell cycle. Kinase activity assays revealed the highest CDK4 and CDK6 activity at G1 phase of the cell cycle (Figure 3d), which correlated well with the highest expression of *CYCLIN D1*, *CYCLIN D3* and *CDK6* transcripts at G1 phase (Figure 3b). *CYCLIN E* protein expression was higher at G1 and S phases compared with its expression at G2 (Figures 3b and c). The increase in the *CYCLIN E* protein level at the G1 and S stages can be explained by its significant phosphorylation at Thr 395 at S and G2 stages, which corresponds to proteolysis and ubiquitin-dependent degradation of the protein (data not shown). The protein expression of *CYCLIN A* and *c-MYC* was higher in S and G2 phases when compared with expression at G1, whereas expression of *CYCLIN B1* was highest at G2 and *CDC25A* at G1 (Figure 3c). It is interesting to note that *CDK2* protein expression is higher in S phase when compared with G1 and G2 and this correlates with maximal expression of the Thr 160 phosphorylated form of *CDK2*, which corresponds to protein activation (Figure 3c) and maximal *CDK2* kinase activity (Figure 3d). Notwithstanding this, the inactivated form of *CDK2* (phosphorylated at Thr 14 and Tyr 15) shows the highest expression at S and G2 phases of the cell cycle. In accordance with *CDK2* and *CYCLIN E* cell cycle-dependent regulation, the expression level of *c-MYC* (Figure 3c) demonstrated a cell cycle-dependent fluctuation too, suggesting its activation as a possible second pathway that can take place at this time. Taken together, these results suggest important differences in cell cycle regulation between murine- and hESCs.

CDK2 knockdown induces changes in cell morphology, cell cycle profile, expression of pluripotency markers and cell cycle inhibitors

The high activity of *CDK2* together with its cell cycle-dependent expression suggests that *CDK2* might play an important role in cell cycle regulation in hESCs. To investigate whether *CDK2* expression is needed to maintain the rapid proliferation and cell cycle regulation of hESCs, we performed transient transfection experiments with two pooled *CDK2* siRNAs. On the second day after transfection, the expression of *CDK2* was downregulated by more than 90% as assessed by quantitative RT–PCR (Figure 4a). In addition, downregulation of *CDK2* was confirmed by western blotting (Figure 5a) and immunocytochemistry (Supplementary Figure 4). In accordance, *CDK2* kinase activity decreased to 37.4% when compared with the control transfected cells (Figure 4b). Downregulation of *CDK2* induced a significant alteration in cell cycle, as majority of the cells (96.9%) were arrested at G1 phase after 48 h post-transfection (Figures 4k and l) and 47.4% of the cells were still residing at G1, 72 h post-transfection (Figure 4m). In addition, changes in cell morphology, cell shape, nuclear–cytoplasmic ratio and adhesiveness were noted during the first 4 days immediately after siRNA transfections (Figures 4e–h). We examined transfected cells for the expression of the SSEA-4 cell surface pluripotency

marker by flow cytometry and revealed that on day 4 after *CDK2* siRNA transfection, the population of SSEA-4-positive cells decreased dramatically (Figure 4o). Normal hESCs morphology was observed 6 days after transfection (Figures 4i and j), whereas a normal cell cycle profile was

also resumed as soon as day 4 after transfection (Figure 4n). It is interesting to note that at day 4 after transfection, 50% of normal *CDK2* expression is achieved (Figure 4a), suggesting that this is perhaps sufficient for return to normal cell cycle distribution.



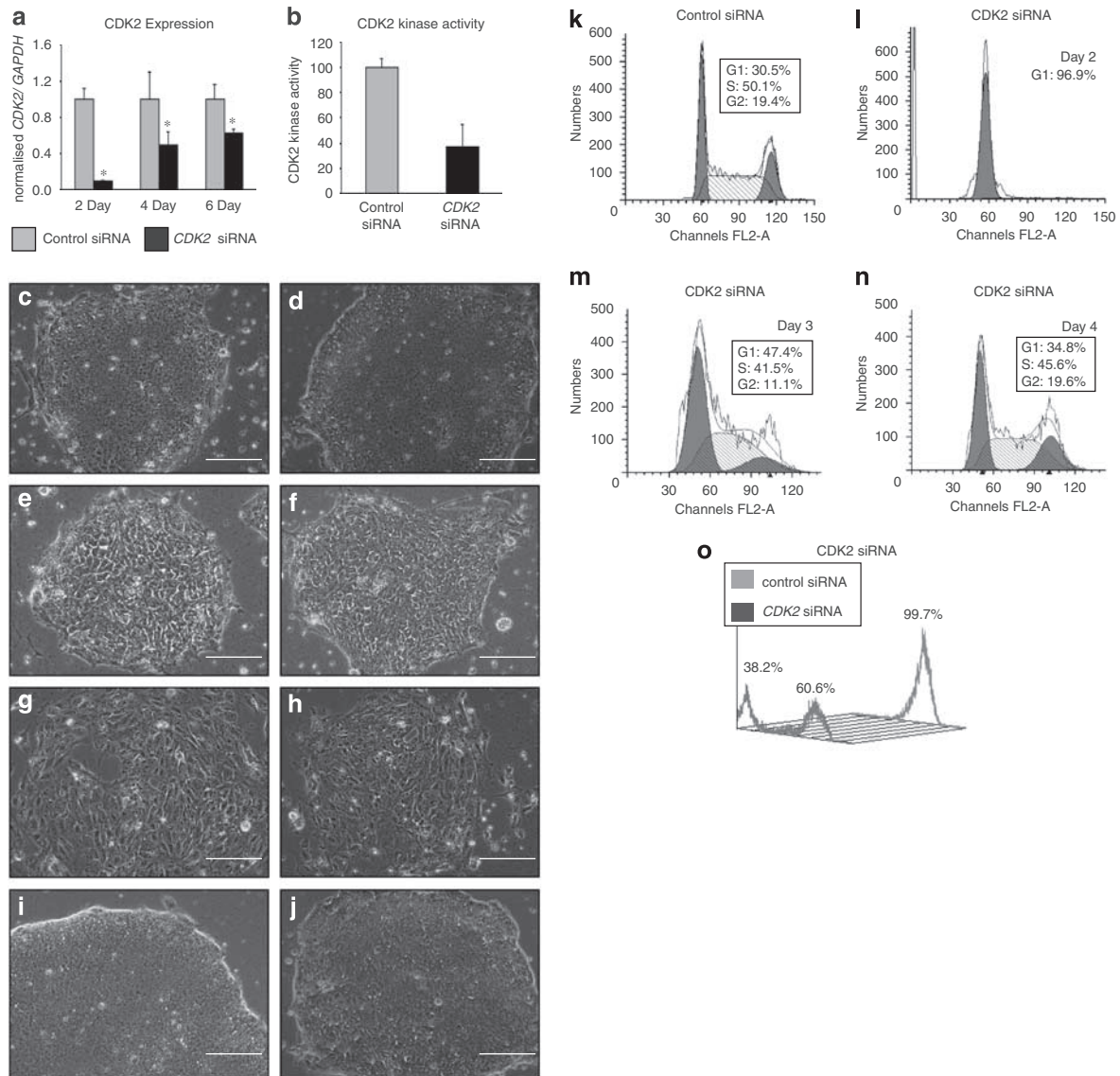


Figure 4 CDK2 downregulation induces changes in cell cycle and loss of pluripotent marker expression in hESCs. **(a)** Downregulation of CDK2 expression at 2, 4 and 6 days after siRNA transfection assessed by quantitative RT-PCR. The data represent the mean $\pm$ s.e.m. from three independent experiments. Statistical significance was assessed using Student's *t*-test, $*P < 0.05$. **(b)** CDK2 kinase activity in hESCs (H9) transfected with control siRNA and CDK2 siRNA on the second day after transfection, shown as percentage change from control. The data represent the mean $\pm$ s.e.m. from three independent experiments. Statistical significance was assessed using Student's *t*-test, $*P < 0.05$. **(c and d)** Bright field microscopy images of hESCs transfected with control siRNA at 48 h post-transfection. **(e and f)** Bright field microscopy images of hESCs transfected with CDK2 siRNA at 48 h post-transfection. **(g and h)** Bright field microscopy images of hESCs transfected with control siRNA at 96 h post-transfection. **(i and j)** Bright field microscopy images of hESCs transfected with control siRNA at 144 h post-transfection. **(c-j)** Scale bar: 100 μ m. **(k-n)** Flow cytometry analysis of cell cycle (PI staining) profile of hESCs (H9) treated with control siRNA **(k)** and H9 cells transfected with CDK2 siRNA 2 days **(l)**, 3 days **(m)** and 4 days **(n)** after transfection. MODFIT analysis was applied to assess cell cycle distribution. A representative example of at least three independent experiments is shown. **(o)** Flow cytometry analysis of the expression of cell surface marker SSEA4 in hESCs 4 days after transfection of control or CDK2 siRNA. hESCs, human embryonic stem cells.

Figure 3 Expression analysis of CYCLINS and CDKS during cell cycle progression. **(a-d)** Cell cycle profile of PI-stained hESCs (H9) by flow cytometry. **(a)** Histogram representation of the cell cycle distribution of unsynchronised cells (control) and hESCs synchronised at G1, S and G2 phases. Histograms show one representative example from three independent experiments. Frequencies of the cells in each phase of the cell cycle were calculated using MODFIT-LT software. **(b)** Quantitative RT-PCR analysis of *CYCLINS*, *CDKS*, *c-MYC* and *CDC25A* in unsynchronised and synchronised hESCs. Results are presented as average $\pm$ s.d. ($n = 3$). Statistical significance was Student's *t*-test ($*P < 0.05$; $**P < 0.005$). **(c)** Lysates (20 μ g protein) from asynchronous and synchronized hESCs and collected at the particular stage of the cell cycle, indicated as G1, S or G2 were subjected to immunoblot analysis to analyse the expression of various CYCLINS and CDKS at different stages of the cell cycle. A representative example of at least three independent experiments is shown. **(d)** Analysis of CDK2, CDK4 and CDK6 kinase-associated activity assays during hESCs cell cycle progression. Results are presented as average $\pm$ s.d. ($n = 3$). Statistical significance was Student's *t*-test ($*P < 0.05$). hESCs, human embryonic stem cells.

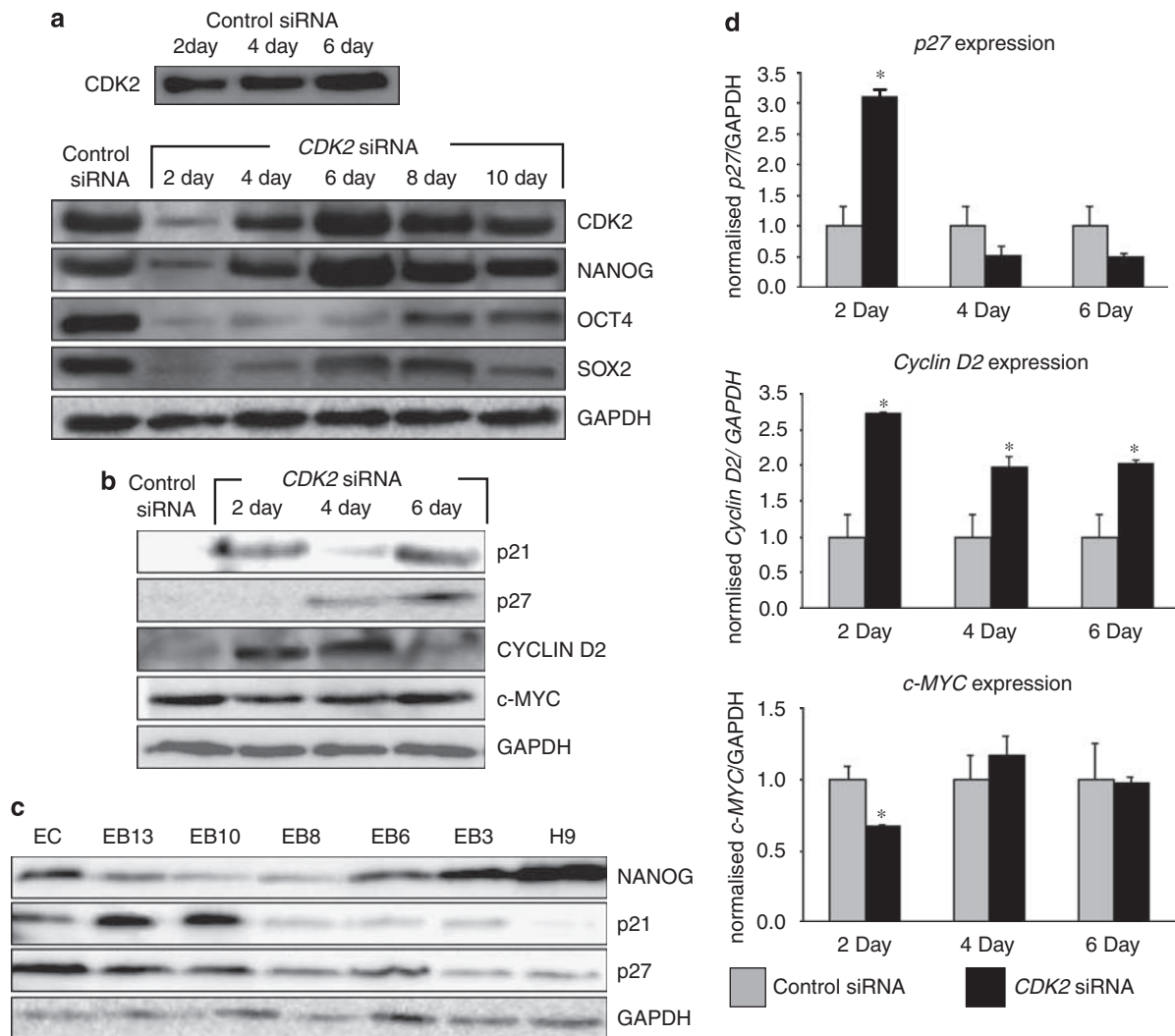


Figure 5 *CDK2* downregulation induces changes in expression of pluripotency markers and cell cycle inhibitory proteins. (**a**-upper panel) Lysates from H9 cells transfected with control siRNA during the course of transient transfection were subjected to immunoblot analysis of *CDK2* expression on days 2, 4 and 6 after siRNA transfection; (**a**-lower panel) Immunoblot analysis of protein expression (15 μ g of protein) of *CDK2*, *SOX2*, *NANOG* and *OCT4* during the course of *CDK2* siRNA transfection. (**b**) Expression of p21, p27, c-MYC and *CYCLIN D2* during the course of *CDK2* siRNA transfection by direct immunoblotting (15 μ g of protein). (**c**) Immunoblot analysis (20 μ g of protein) of expression of pluripotency marker *NANOG* and differentiation markers p21 and p27 during hESCs differentiation using the EB method. EBs 3, 6, 8, 10 and 13 at days 3, 6, 8, 10 and 13 of differentiation. GAPDH was used as loading control. In all panels a representative example of at least three independent experiments is shown. (**d**) Quantitative RT-PCR for expression of p27, *CYCLIN D2* and c-MYC in *CDK2* and control siRNA-transfected cells. Results are presented as average $\pm$ s.d. ($n=3$). Statistical significance was Student's *t*-test (* $P<0.05$). EB, embryoid body; GAPDH, glyceraldehyde-3-phosphate dehydrogenase; hESCs, human embryonic stem cells.

The observed prominent morphological changes in *CDK2* siRNA-transfected cells, loss of hESCs morphology and SSEA-4 expression (Figures 4e–h and o) suggested that *CDK2* knockdown along with downregulation of *CDK2* kinase activity might affect hESCs pluripotency. Using direct immunoblotting, we observed downregulation of important hESCs markers: *OCT4*, *SOX2* and *NANOG* at days 2 and 4 after transfection (Figure 5a). As the effects of RNA-mediated interference are transient, upregulation of *CDK2* and *NANOG* expression was observed from day 4 by western blotting (Figure 5a). An increase was also observed in the expression of *OCT4* (albeit still lower

than normal when compared with control siRNA-transfected sample) by day 8 post-transfection (Figure 5a), while 50% of *SOX2* expression was acquired at day 6, suggestive of different kinetics of regulation for these three pluripotent markers. It is important to note that transfection of hESCs with nucleofector solution alone did not result in changes in gene expression or cell cycle distribution when compared with hESCs transfected with control siRNA (Supplementary Figure 5), thus suggesting that the effects we have observed on cell cycle regulation and differentiation are linked to downregulation of *CDK2* specifically.

CYCLIN D2 expression was significantly increased at both mRNA and protein level (Figures 5b and d) from the second day after transfection, suggesting the start of differentiation process in hESCs (compare with Figure 1b). Quantitative RT-PCR analysis with lineage-specific markers indicated a significant increase in expression of *IHH* (primitive endoderm marker), *GATA6* (primitive endoderm marker), *CDX2* (trophoblast marker), *FG5* (primitive ectodermal marker) at 6 days post-transfection consistent with reduced expression of OCT4 and SOX2 also observed at the same day; however, no significant changes in expression were observed for *PAX6* (neuroepithelial marker) or *BRACHYURY* (mesodermal marker, Supplementary Figure 6). These data indicate initiation of hESCs differentiation to extraembryonic lineages upon CDK2 knockdown. Acquisition of normal hESCs morphology and normal cell cycle distribution observed in Figure 4 might lead to speculations that differentiation process caused by CDK2 knockdown is reversible. This however is not supported by reduced expression of OCT4 and SOX2 even at 10 days post-transfection and upregulation of extraembryonic lineage markers seen at day 6 (in the presence of full CDK2 expression). It is known that OCT4, SOX2 and NANOG cooperate closely together to maintain hESCs pluripotency (Boyer *et al.*, 2005) and downregulation of OCT4 on its own causes differentiation of hESCs to extraembryonic lineages (Zachres *et al.*, 2005), similarly to what has been observed in this study (Supplementary Figure 6). Taken together, these data point to non-reversible effects on differentiation; however, it has to be stressed that this can be addressed in much more detail in homogenous cell populations obtained after transfection of inducible hairpin siRNAs.

We investigated whether the observed downregulation of pluripotency markers and alteration in the cell cycle structure and CDK2 kinase activity on downregulation of *CDK2* would induce the expression of differentiation markers, such as p27 and p21, thus providing a link between cell cycle regulation, pluripotency and self-renewal. Western blot analysis of CDK2 siRNA-transfected cells demonstrated a marked increase in the expression levels of p21 from the second day after transfection and p27 by day 4 (Figure 5b). This was in agreement with the increased protein levels of p27 and p21 observed in hESCs during their differentiation by day 10 (Figure 5c). Quantitative RT-PCR analysis confirmed upregulation of *p27* mRNA on the second day post-transfection with *CDK2* siRNA (Figure 5d). Importantly, we observed downregulation of *c-MYC* (Figures 5b and d) expression on day 2 after transfection, but no significant changes in the level of *CYCLIN E*, *A* and *CDC25A* were detected (data not shown).

Discussion

During cell cycle progression, a cell may adopt one of five different fates: it can proliferate, differentiate, become quiescent, senescent or enter apoptosis. The

fate decision is often made in the G1 phase of the cell cycle in which the cell may utilize an early checkpoint that can determine the cell's fate choice upon detection of DNA damage or stress stimuli (Blomen and Boonstra, 2007).

In contrast to mouse embryonic stem cells, which are characterized by a higher CDK6 expression and lack of CDK4 activity (Faast *et al.*, 2004), our investigation of hESCs demonstrated that expression of CDK4 is higher than that of CDK6, in agreement with an earlier publication (Becker *et al.*, 2006). In addition, CDK2 kinase activity is higher than that of CDK4 and CDK6, which may reflect the formation and a turnover of several complexes such as CDK2-CYCLIN E, CDK2-CYCLIN A, CDK2-CDC25A and CDK2-c-MYC as revealed by our immunoprecipitation studies. We also noticed significant differences in the expression of these cell cycle components between hESCs and EC cells, although this may result from higher gene copy numbers as a result of the aneuploidy associated with EC cells (Wang *et al.*, 1980). Many EC cell lines show reduced tendency to undergo spontaneous differentiation when compared with hESCs, suggesting that they have been subjected to strong selections for culture adaptations/mutations that lead to an increase in self-renewal compared with differentiation (Andrews *et al.*, 2005). Higher expression, therefore, of some cell cycle regulatory components (for example, c-MYC, CDK2, CYCLIN D1) might be responsible for tuning the fine balance between self-renewal and differentiation, and needs to be investigated in detail in hESCs and during their culture adaptation (Enver *et al.*, 2005).

We examined the protein expression pattern of the main cyclin-dependent kinases during cell cycle progression using a combination of quantitative RT-PCR and direct immunoblotting, which indicated that most of the cell cycle components in hESCs show cell cycle-dependent expression. Importantly, the active form of CDK2 is most highly expressed in S phase and cyclin E reaches its highest expression during G1 and S phases, whereas CYCLIN A is upregulated during S and G2 phases of the cell cycle. At the transcriptional level *CDK6*, *CYCLIN D1* and *CYCLIN D3* are also upregulated at G1, which is confirmed by kinase activity assays that revealed highest CDK6 and CDK4 activity in G1 and CDK2 activity in S phase. For the first time, our studies showed that CDK2, CDK4 and CDK6 activity is also regulated in a cell cycle-dependent manner in hESCs in contrast to mouse embryonic stem cells, which show high and constant Cdk2 activity throughout cell cycle progression and undetectable Cdk4 activity. One speculative interpretation of these findings that takes into account the immediate upregulation of Cdk/cyclin D activity in murine ESCs on their differentiation would be that hESCs are more similar to early differentiated murine ESCs, where the more sophisticated cell cycle-dependent Cdk/cyclin D mechanisms are established. Indeed, new studies have shown that hESCs are more similar to the pluripotent cell lines derived from the post-implantation murine epiblast (Brons *et al.*, 2007; Tesar *et al.*, 2007).

Interestingly, inhibition of Cdk2/Cyclin E activity in murine ESCs using a pharmacological inhibitor results in lengthening of the cell cycle over all phases consistent with the cell cycle-independent expression of Cdk2 and Cyclin E, without impact on expression of pluripotent markers such as Oct4 or Rex1 (Stead *et al.*, 2002). CDK2 showed the highest kinase activity compared with other G1-specific CDKs (CDK4 and CDK6); we, therefore, hypothesised that CDK2 might play an important role in maintaining hESCs proliferation and pluripotency, and set out to investigate its function. Transfection with *CDK2* siRNAs induced remarkable changes in cell morphology and cell cycle arrest at the G1 stage. This was accompanied by the downregulation of specific pluripotency markers such as OCT4, SOX2 and NANOG, as well as cell surface marker SSEA-4 and differentiation of hESCs to extraembryonic lineages. At the same time, we observed a significant increase in the expression of CDK inhibitors, p27 and p21, known to play an important role in induction of differentiation.

In summary, results of the present study show the presence of the most common cyclins and Cdk2s required for G1 to S phase transition, and point to an important role for CDK2 in the G1 to S phase transition in hESCs. Most importantly, these studies reveal for the first time some important differences in the regulation of expression and activity of G1 to S cell cycle regulatory components between human and murine ESCs, which has not been described earlier. Finally, our data suggest some intrinsic link between cell cycle regulation and cell-fate decisions in hESCs (self-renewal versus differentiation) that are likely to be affected by the length of each phase of the cell cycle.

Materials and methods

Culture and differentiation of hESCs

Human H1 and H9 (WiCell Research Institute, Madison, MI, USA) hESC lines were used in this study. hESCs were grown on mitotically inactivated mouse embryonic fibroblasts and passaged essentially as described in Stojkovic *et al.* (2004a, b). Embryoid body differentiation was induced as described in Stojkovic *et al.* (2004a, b). A few passages prior to their use in experiments, hESCs were transferred to Matrigel-coated plates with feeder conditioned media as previously described in Stojkovic *et al.* (2004a, b). Our studies have indicated that there are no significant changes in cell cycle structure between the feeder-free and feeder-based hESCs culture (data not shown). Karyotypic analysis was carried out for every 15 passages, and in all cases normal karyotype was observed for hESCs used in this study (Supplementary Figure 1).

Small interfering RNAs and transient transfection

CDK2 siRNAs were obtained from Invitrogen Ltd (Paisley, UK) (Validated Stealth RNAi DuoPak for CDK2 NM_001798.2; duplex 1: 5'-CCUAAAACCUCAGAAUCU GCUAAUU-3' and duplex 2: 5'-CCUAUUCUGGAGA UUCUGAGAUU-3'). Transfection with a mixed pool of scrambled control siRNAs that had the same GC content as the CDK2 siRNAs provided by Invitrogen was used as a negative control. Transfection of siRNA into small clumps of

hESCs was carried out using the high-efficiency nucleofection kit L from Amaxa AG (Cologne, Germany) (program A-023) and 80 pmol siRNA (in 2 ml medium) as outlined in manufacturer's instructions. The transfected cells were plated on Matrigel-coated plates in the presence of feeder-conditioned media as described above. The cells were analyzed 48, 72 and 96 h after transfection.

Flow cytometry analysis of hESCs

For the flow cytometry analysis the hESCs were collected, processed and analyzed as described in Armstrong *et al.* (2006) with the following antibodies: rabbit anti-cyclin D1 monoclonal antibody (FITC conjugated, clone SP4 from Abcam plc, Cambridge, UK); mouse monoclonal anti-Cyclin E (HE 12; FITC conjugated from Santa Cruz Biotechnology Inc., Santa Cruz, CA, USA) and R&D systems (Minneapolis, MN, USA) monoclonal anti-human/mouse SSEA-4 (APC conjugated).

Cell cycle analysis

Cell cycle analysis was performed using the CycleTest Plus DNA reagent kit (Becton Dickinson, San Jose, CA, USA). hESCs were harvested by accutase treatment and counted (hemocytometer). In total, 500 000 cells were fixed, permeabilized and stained in accordance with the manufacturers' instructions and the sample was analyzed by flow cytometry (Becton Dickinson FACS Calibur) measuring FL2 area versus total counts. The data were analyzed using ModFit (Verity Software House, USA, www.vsh.com) to generate percentages of cells in G1, S and G2/M phases.

Synchronization experiments

On day 3 after plating, hESCs cultured on Matrigel-coated plates were synchronized at the G2 stage of the cell cycle by incubation for 16 h with 200 ng/ml nocodazole (Sigma-Aldrich Ltd, Dorset, UK) as described by Becker *et al.* (2006). For collection of cells at the G1 stage of the cycle, cells were first treated by Nocodazole for 16 h, then the drug was removed by several washes with media and a fresh media supplemented with 10 µg/ml of aphidicholine (Sigma) was applied for the next 9–10 h as it was reported for mouse embryonic stem cells (Stead *et al.*, 2002). To demonstrate that cells were properly released from blocking procedures, cell cycle progression was analysed after release from inhibitors (Supplementary Figure 2) at every 1–2 h. For example, cells that were blocked in G1 by application of nocodazole–aphidicholine block took 4–5 h to resume the normal cell cycle distribution (Supplementary Figure 2).

Western blotting

Human ESCs were washed with ice-cold PBS and lysed in RIPA buffer (50 mM Tris-HCl pH 8.0, 150 mM NaCl, 1% IGEPAL CA 630, 0.5% Na-DOC and 0.1% SDS). One millimolar phenylmethanesulphonylfluoride (PMSF) and Roche protease inhibitors were added before treatment of cells. After 30 min on ice, the lysates were homogenized and centrifuged 13 000 r.p.m. 15 min. The total protein concentration was determined by Bradford Kit (Bio-Rad Laboratories Ltd, Hemel Hempstead, UK) using the manufacturer's instructions. Lysates (10–20 µg total protein) were electrophoresed on a 10–12% SDS-PAGE gel and electrophoretically transferred to a polyvinylidene difluoride membrane (Hybond-P (hydrophobic polyvinylidene difluoride membrane, cat no. RPN303F); Amersham Biosciences, Piscataway, NJ, USA). Membranes were blocked in Tris-buffered saline with 5% milk and 0.1% Tween. The blots were probed with anti-NANOG (Aviva Antibody Corporation – AVIVA Systems Biology, CA, USA),

SOX2 (R&D), CDK4, CDK6, CDK2, CDK1 (Santa Cruz Biotech), CYCLIN D1, D2, D3, CYCLIN E, CYCLIN A and B1, p21, p27, OCT4 and β -Actin (all from Santa Cruz Biotech), CDC25A and c-Myc (from Cell Signalling Technology, Inc., MA, USA) and GAPDH antibody (Abcam) overnight and revealed with horseradish peroxidase-conjugated secondary anti-rabbit or anti-mouse antibodies (Amersham Biosciences). Antibody-antigen complexes were detected using ECL Plus reagent (Amersham Biosciences) and GeneSnap software (version 4.00.00; Syngene Ltd, Cambridge, UK) with GeneGnome (Syngene). Bands were analysed using the GeneTools software (version 3.00.22; Syngene). Antibodies to β -actin or GAPDH were used after membrane stripping to confirm uniform protein loading.

Immunoprecipitation studies

Immunoprecipitations were performed with 2 μ g of polyclonal CDK6 antisera (C-21; Santa Cruz Biotechnology), polyclonal anti-CDK4 (C-22; Santa Cruz Biotechnology), monoclonal anti-CYCLIN D1 (DCS-6; Santa Cruz Biotechnology), monoclonal anti-CYCLIN D3 (D7; Santa Cruz Biotechnology), polyclonal anti-CYCLIN E (M20; Santa Cruz Biotechnology), monoclonal anti-CDK2 (D-12; Santa Cruz Biotechnology) and monoclonal anti-CDC25A (F6; Santa Cruz Biotechnology) antibodies. hESCs were lysed in RIPA lysis buffer and 400–500 μ g of extract was immunoprecipitated for at least 2 h at 4 °C with mixing. In total, 40 μ l of swollen protein A/G-Sepharose beads were added for an additional 2–3 h, washed three times with complete RIPA lysis buffer and separated on denaturing acrylamide gels. The following antibodies: CDK4 (DCS-35; Santa Cruz Biotechnology), CDK6 (B-10; Santa Cruz Biotechnology), CYCLIN D1 (DCS-6; Santa Cruz Biotechnology), CYCLIN D2 (C-17; Santa Cruz Biotechnology), CYCLIN D3 (D7; Santa Cruz Biotechnology), CYCLIN E (HE12; Santa Cruz Biotechnology), CDK2 (M2; Santa Cruz Biotechnology), CYCLIN A (H-432; Santa Cruz Biotechnology), c-MYC (9E10; Santa Cruz Biotechnology) and CDC25A (3652; Cell Signalling) were used for subsequent western blotting analysis.

Kinase activity assays

Kinase activity assays were carried out using the PKLight Assay Kit (LT07-500) from Cambrex Bio Science Rockland Inc., ME, USA following the manufacturer's instructions. The PKLight Assay exploits the kinases' intrinsic ATPase activity, resulting in the cleavage of the γ -phosphate moiety of ATP and its subsequent insertion into the target substrate. This results in the phosphorylation of the substrate and the conversion of ATP to ADP. The PKLight Assay measures the consumption of ATP and is based on the bioluminescent measurement of the remaining ATP present in the wells after the kinase reaction. Bioluminescent signal of PKLight Assay is inversely proportional to kinase activity. Phosphorylation of RB or H1 was measured by incubating for 10 min at room temperature 20 μ l of immunoprecipitation product for the kinase of interest (see above) with 1 mM ATP, kinase buffer (50 mM Tris, pH 7.5, 5 mM MgCl₂) and RB or H1 (5 mg/ml) as a substrate. In total, 10 μ l of kinase stop solution was added to each sample at room temperature for 10 min. Finally, 20 μ l of ATP detection reagent was added to each sample at room temperature for 10 min and

the readings were taken using a luminometer. The difference in luminometer reading between the no antibody control and immunoprecipitated product containing the antibody was calculated. This figure, which is indicative of remaining ATP in the solution, was inversely correlated to the kinase activity.

Isolation of RNA and quantitative RT-PCR analysis

Total RNA was prepared using TRIzol reagent (Invitrogen Corp.) as recommended by the manufacturer. After being isolated, treated with DNaseII, quantified and total RNA was reverse-transcribed to cDNA using random primers (Promega Ltd, UK). The reaction mixture, containing SYBR Green PCR Master Mix (Sigma), was run in a 7900HT Fast Real-Time PCR System (Applied Biosystems). The sequences of the oligonucleotides used for the quantitative RT-PCR are shown in Supplementary Table 1. Regression curves were drawn for each sample and the relative amount was calculated from the threshold cycles with the instrument's software (SDS2-3 software: Applied Biosystems, Foster City, USA) according to the manufacturer's instructions. Relative expression levels of the target genes were normalized with the control gene *GAPDH*.

Immunocytochemistry

Cells grown on Matrigel-covered plastic plates were quickly washed with phosphate-buffered saline (PBS), fixed with cold methanol for 10 min on ice and permeabilised with 0.25% Triton X-100 in PBS for 15 min at room temperature. Unspecific binding was blocked by incubation of samples in PBS containing 3% FCS for 40 min. Primary antibodies (mouse monoclonal anti-CDK2, D-12 from Santa Cruz) were applied overnight at constant agitation at 4 °C. After 30 min wash with PBS (three times for 10 min), the secondary Abs (anti-mouse FITC-conjugated from Sigma) were applied for 2 h at room temperature. Then samples were washed for 30 min in PBS solution containing 0.1% Tween and cells were stained with DAPI (4',6-diamidino-2-phenylindole) nuclear stain for 10 min at room temperature in the dark. After 30 min PBS wash, samples were covered with Vectashield Mounting Medium (Vector Laboratories Ltd, Peterborough, UK) and analysed under a fluorescence microscope (Axiovision Viewer Software 4.6: Carl Zeiss Jenna, GmbH, Gottingen, Germany).

Statistical analysis

Two-tailed pairwise Student's *t*-test was used to analyse the results obtained from two samples with one time point. Anova two factors were used to analyse results obtained from multiple time points. The results were considered significant if *P* < 0.05.

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Supplementary Information accompanies the paper on the Oncogene website (<http://www.nature.com/onc>)